

Wyatt Walsh

SENIOR AI/ML ENGINEER

wyattowalsh@gmail.com (209) 602-2545 New York City, New York [linkedin.com/in/wyattowalsh](https://www.linkedin.com/in/wyattowalsh) github.com/wyattowalsh

Senior AI/ML engineer shipping production agentic AI, LLM document intelligence, retrieval workflows, data pipelines, and fintech developer tooling.

EXPERIENCE

Senior AI/ML Engineer

Frame Payments | Remote | Oct 2025 - Jan 2026

- Deployed production LLM document intelligence with PydanticAI, Amazon Bedrock AgentCore, and Claude; added human-in-the-loop browser feedback for confidential uploads and reduced analyst follow-ups/resubmissions ~90% by blocking miscategorized documents before review.
- Built multimodal restricted-item compliance screening by combining product images and structured metadata; translated banking-partner policy requirements into automated risk checks for merchant onboarding across tens of merchants and multi-location catalogs.

Associate III, Senior Software Engineer - Data Integrations

JPMorgan Chase & Co. | New York, NY | May 2023 - Apr 2025

- Pioneered firm-wide rollout of secure generative AI developer tools (LLMs, GitHub Copilot, Codeium) as one of the first 500 engineers, shaping scalable deployment and leading workshops for technical and non-technical teams.
- Devised ML-driven anomaly detection for credit trading and risk analytics, reducing false positives by 25% and manual review time for risk teams.
- Developed a resilient Python wrapper for KDB+ with CI-style tests and optimized Q functions, democratizing data access for 5,000+ users.

Senior Software Engineer

Stealth Web3 Startup | Remote | Mar 2022 - Jul 2022

- Architected a scalable collateralized lending app with React, Next.js, TypeScript, MobX, and ethers.js for secure wallet and smart-contract workflows.
- Implemented CI/CD with GitHub Actions, Husky, Cypress, and Jest, halving release cycles, raising test coverage to 90%, and mentoring a front-end intern.

Founder, Chief Technology Officer

SandLabs | Remote | Jun 2021 - Mar 2022

- Launched a Next.js/TailwindCSS public site with top-tier Lighthouse scores and secured \$75K in competitive ecosystem grant funding.
- Built Python pipelines that normalized a 1M+ Reddit posts/comments dataset into a reusable analytics data product for external users.

SKILLS

AI, LLM & AGENT ENGINEERING: OpenAI API, Anthropic Claude API, Amazon Bedrock, LangChain, LangGraph, DeepAgents, PydanticAI, Model Context Protocol (MCP), AI Agent Skills, Retrieval Augmented Generation (RAG), Prompt Engineering, Context Engineering, LLM Evaluation, Guardrails AI

DATA ENGINEERING, DATABASES & ANALYTICS: PostgreSQL, KDB+, qPython, DuckDB, Apache Spark, Pandera, Polars, Parquet

STREAMING, MESSAGING & CAPITAL MARKETS SYSTEMS: AMPS, Publish/Subscribe Architecture, Apache Kafka, Amazon Kinesis

CLOUD, DEVOPS & OBSERVABILITY: Docker, Kubernetes, Terraform, AWS Lambda, GitHub Actions

LANGUAGES & AUTHORING: Python, SQL, TypeScript, Go, Rust, q, Bash

PROJECTS

- AI Agent Harness Configs** - Python/TypeScript agent tooling catalog with portable skills, reusable agents, and MCP presets.
- Standardized reusable agent definitions across Claude Code, Codex, GitHub Copilot, Cursor, OpenCode, and related IDE agents.
 - Shipped generated docs, install flows, and MCP-ready tool presets spanning code review, security, infrastructure, research, and workflow automation.

- NBA Basketball Database** - NBA Kaggle dataset/repo with multi-format game/play-by-play exports; rebuilding nbadb around broader extractors and schemas.
- Implemented Pandera-validated NBA API ETL with daily warehouse updates plus DuckDB, SQLite, Parquet, and CSV exports.
 - Building the next-generation nbadb rebuild with broad extractor coverage and generated public star-schema outputs across docs, schema references, and release workflows.

- ProxyWhirl** - Python and SQLite proxy toolkit with 9 rotation strategies, 100+ sources, and self-healing circuit breakers.
- Implemented 9 proxy rotation strategies with configurable fallback chains, per-source scoring, and performance-based selection.
 - Built self-healing validation across 100+ proxy sources with circuit breakers, auto-ejection, recovery promotion, and five interfaces for different scraping workflows.

- Personal Website: w4w.dev** - Next.js and MDX publishing system with polished public UI, durable content workflows, and strong SEO.
- Maintained the public site across multiple generations, covering responsive landing pages, themes, motion, accessibility, and MDX publishing.
 - Built custom MDX publishing with search, metadata, math rendering, feeds, tags, and integrated authoring tools.

EDUCATION & CERTIFICATIONS

Bachelor of Science

University of California, Berkeley - College of Engineering, Industrial Engineering and Operations Research (IEOR)
Aug 2014 - Dec 2020

High School Diploma

The Hotchkiss School
Aug 2012 - May 2014

AWS Certified Cloud Practitioner

Amazon Web Services (AWS) | Nov 2024

IBM Data Science Professional Certificate

IBM | Feb 2021